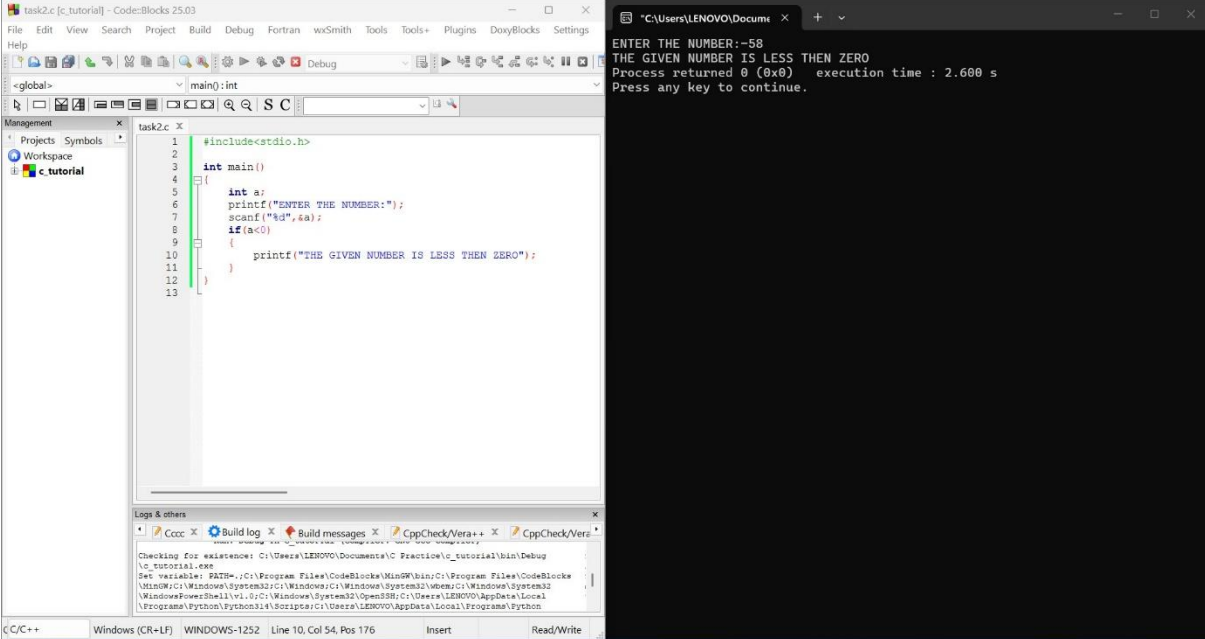


CONTROL STATEMENT: IF TASKS



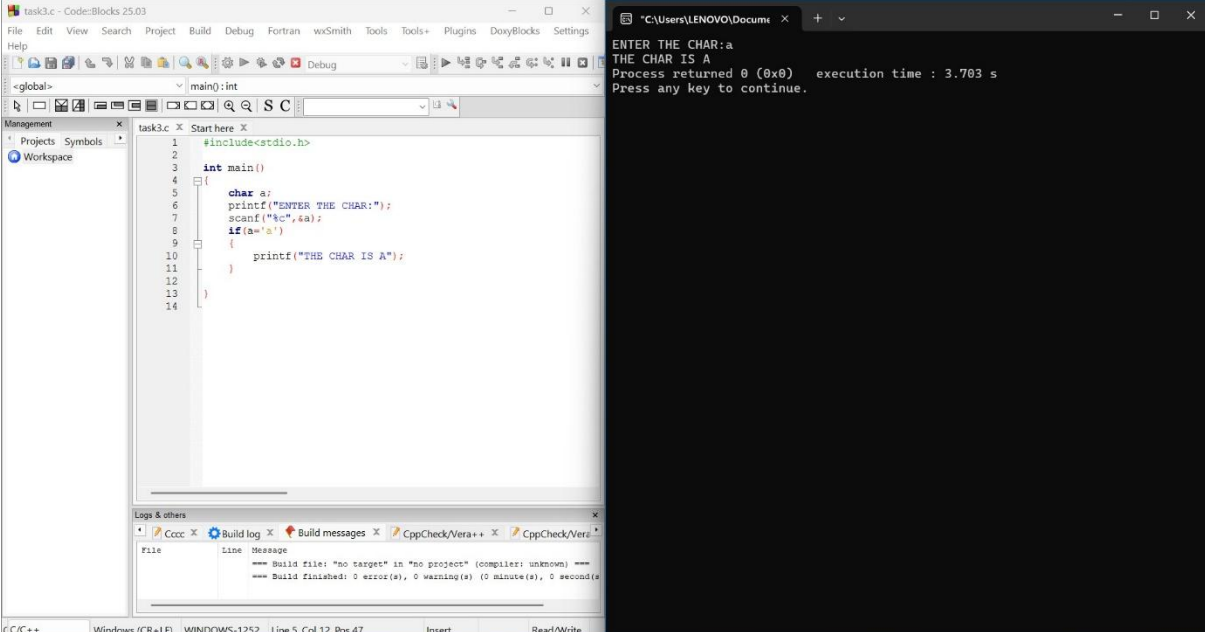
The screenshot shows the Code::Blocks IDE with a project named 'task2.c'. The code in 'main()' is as follows:

```
#include<stdio.h>

int main()
{
    int a;
    printf("ENTER THE NUMBER:");
    scanf("%d",&a);
    if(a<0)
    {
        printf("THE GIVEN NUMBER IS LESS THEN ZERO");
    }
}
```

The execution output in the terminal window is:

```
ENTER THE NUMBER:-58
THE GIVEN NUMBER IS LESS THEN ZERO
Process returned 0 (0x0)   execution time : 2.600 s
Press any key to continue.
```



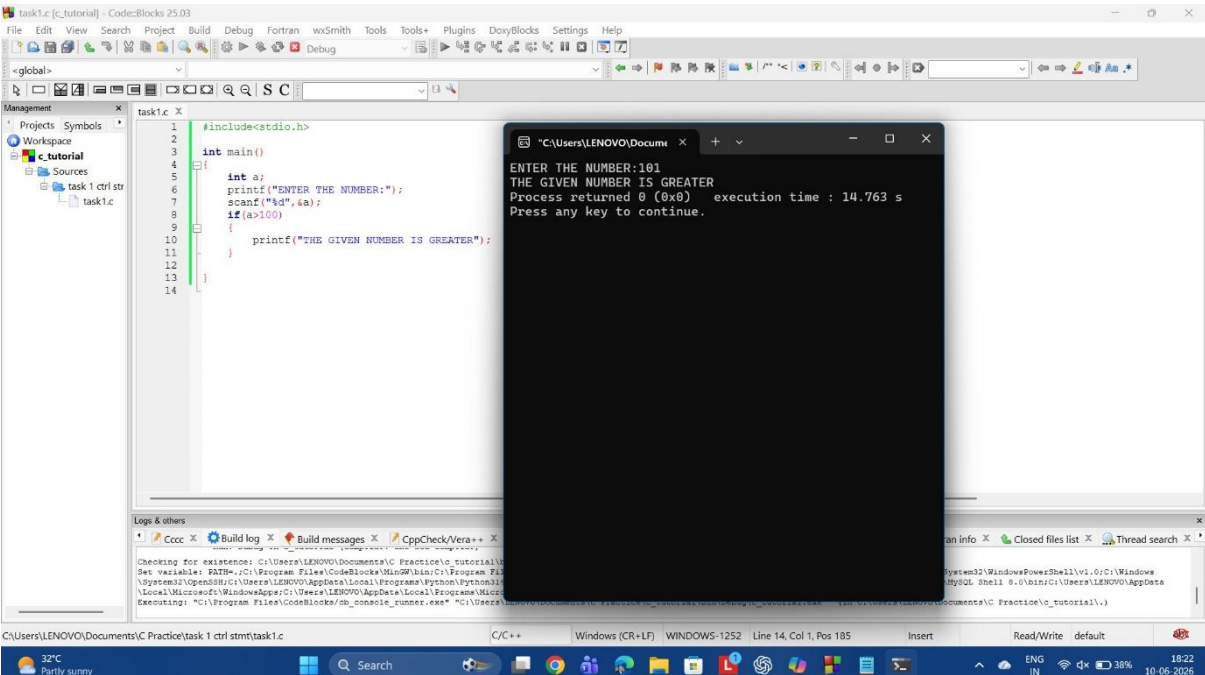
The screenshot shows the Code::Blocks IDE with a project named 'task3.c'. The code in 'main()' is as follows:

```
#include<stdio.h>

int main()
{
    char a;
    printf("ENTER THE CHAR:");
    scanf("%c",&a);
    if(a=='a')
    {
        printf("THE CHAR IS A");
    }
}
```

The execution output in the terminal window is:

```
ENTER THE CHAR:a
THE CHAR IS A
Process returned 0 (0x0)   execution time : 3.703 s
Press any key to continue.
```



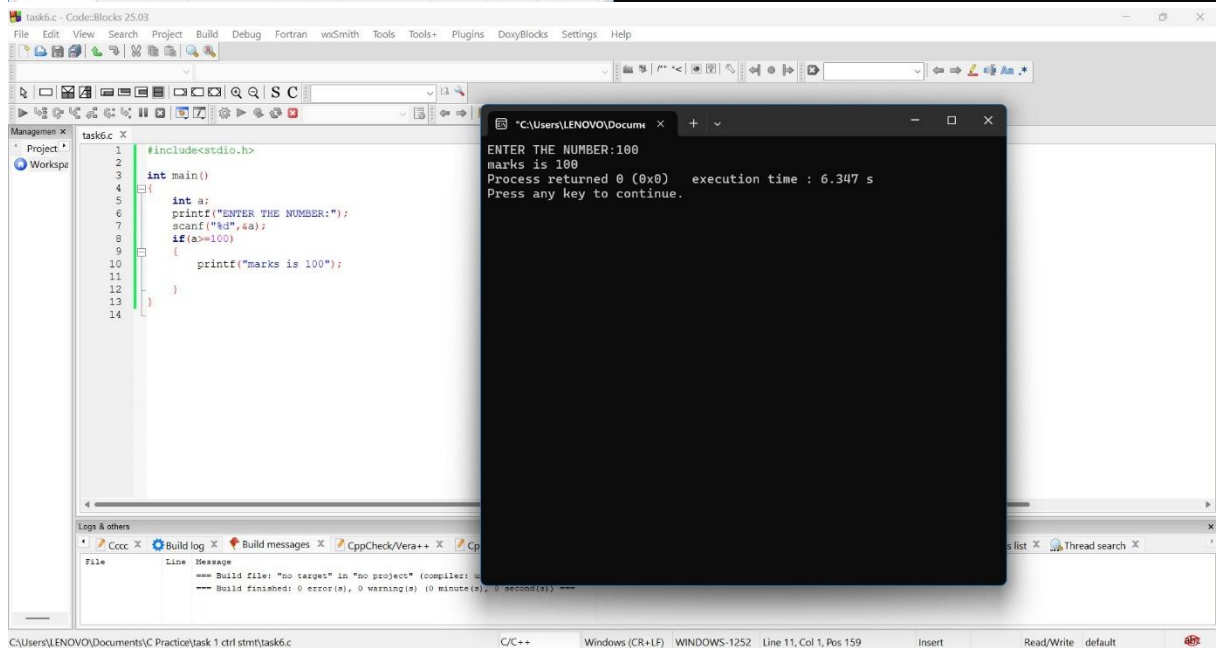
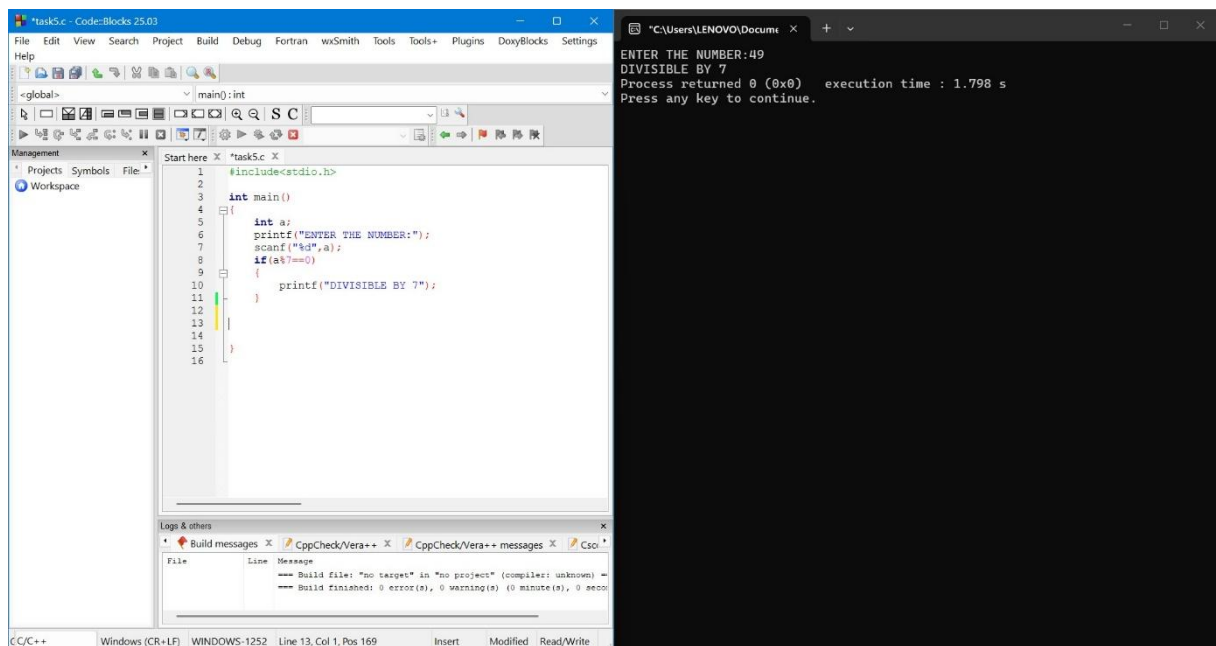
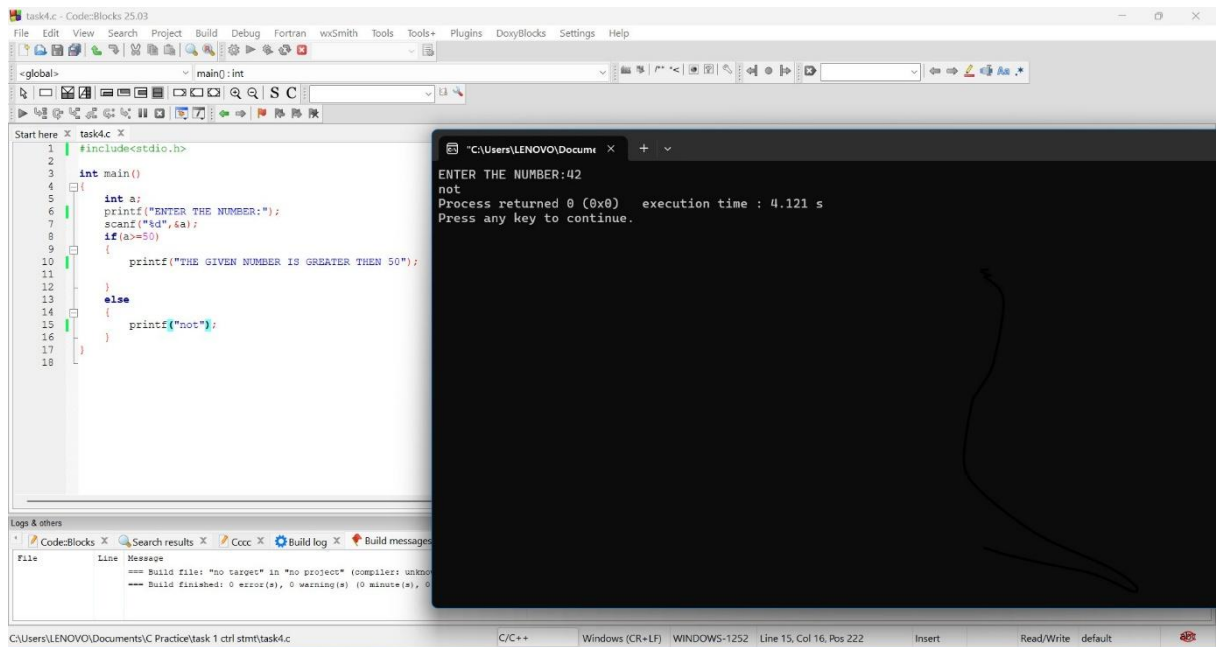
The screenshot shows the Code::Blocks IDE with a project named 'task1.c'. The code in 'main()' is as follows:

```
#include<stdio.h>

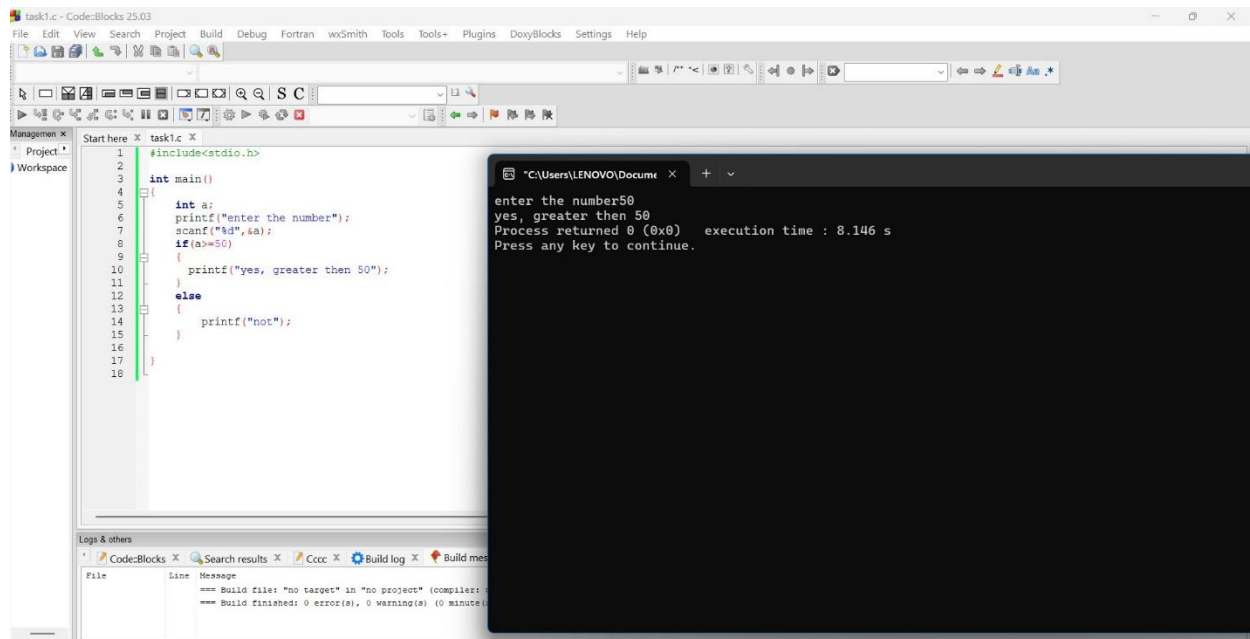
int main()
{
    int a;
    printf("ENTER THE NUMBER:");
    scanf("%d",&a);
    if(a>100)
    {
        printf("THE GIVEN NUMBER IS GREATER");
    }
}
```

The execution output in the terminal window is:

```
ENTER THE NUMBER:101
THE GIVEN NUMBER IS GREATER
Process returned 0 (0x0)   execution time : 14.763 s
Press any key to continue.
```



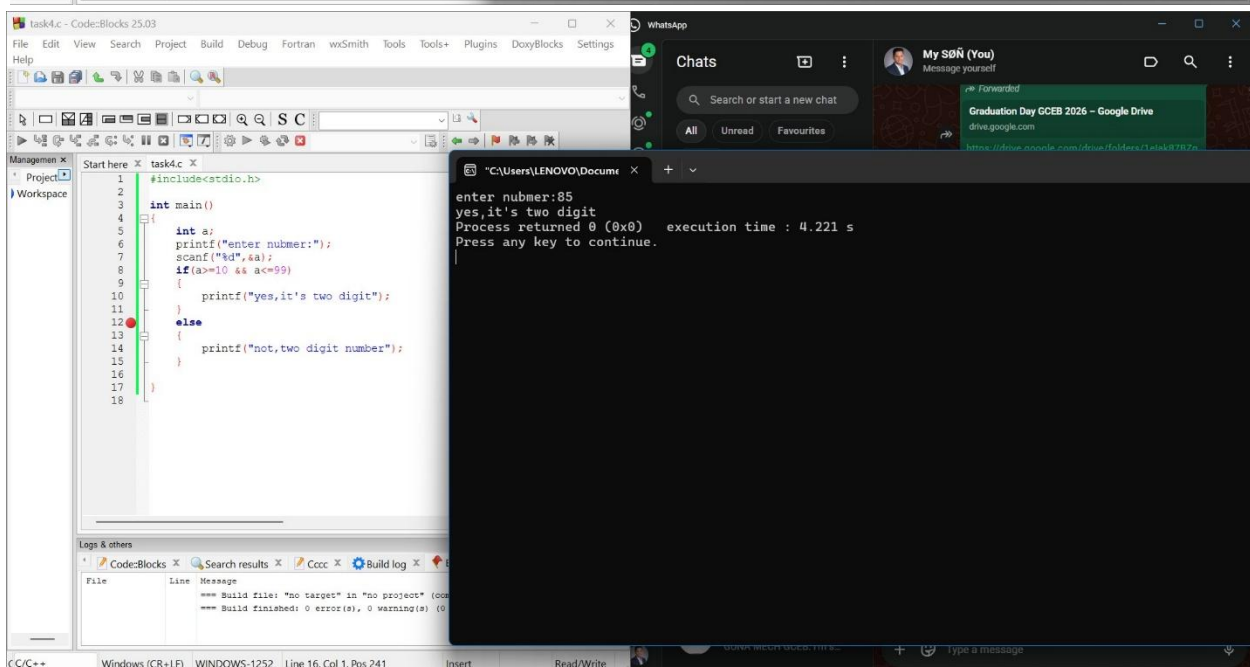
IF-ELSE:



```
#include<stdio.h>

int main()
{
    int a;
    printf("enter the number");
    scanf("%d",&a);
    if(a>=50)
    {
        printf("yes, greater then 50");
    }
    else
    {
        printf("not");
    }
}
```

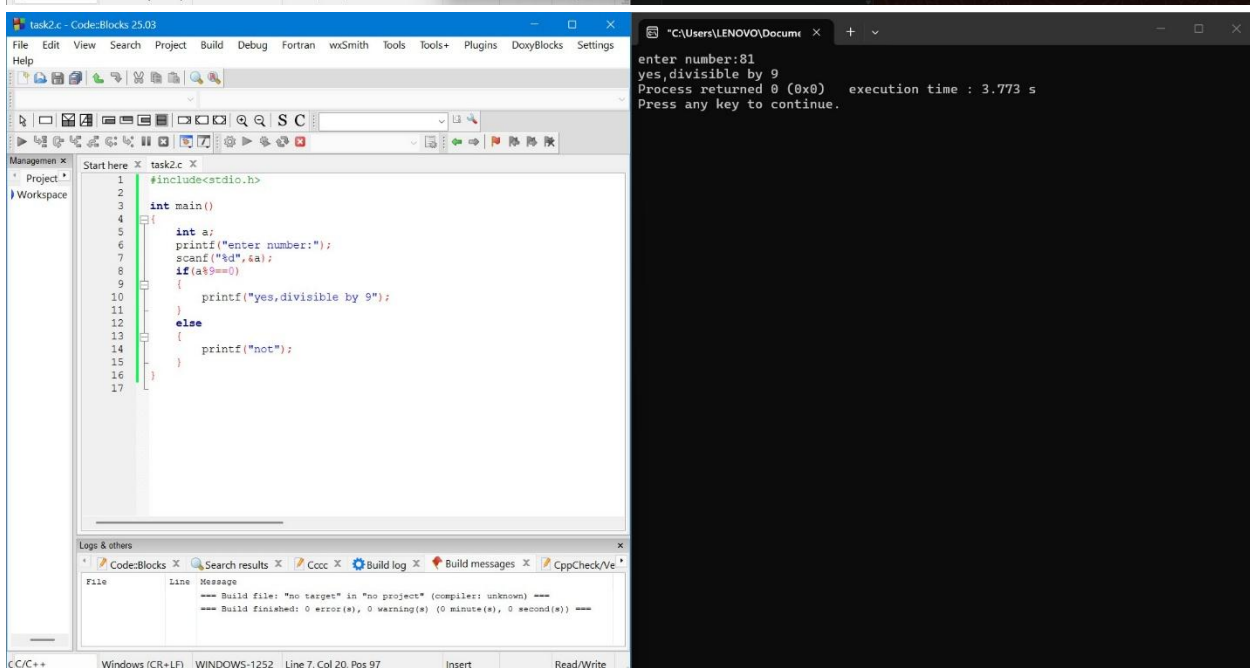
enter the number50
yes, greater then 50
Process returned 0 (0x0) execution time : 8.146 s
Press any key to continue.



```
#include<stdio.h>

int main()
{
    int a;
    printf("enter number:");
    scanf("%d",&a);
    if(a>=10 && a<=99)
    {
        printf("yes, it's two digit");
    }
    else
    {
        printf("not, two digit number");
    }
}
```

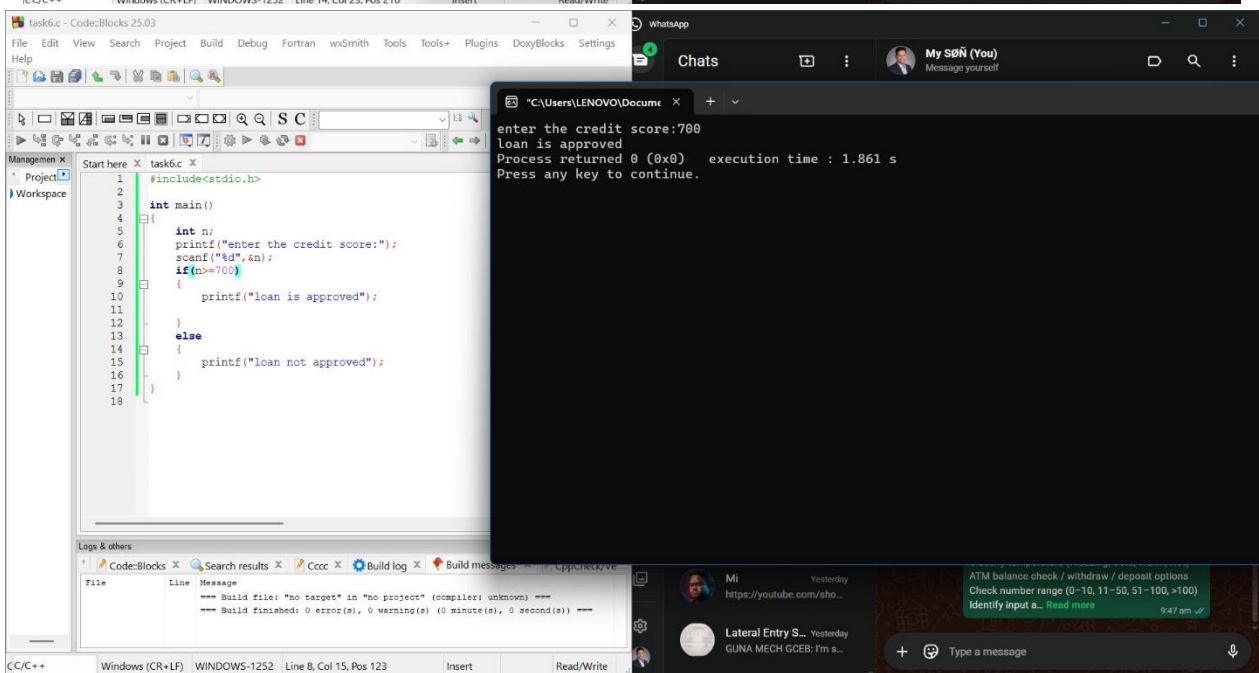
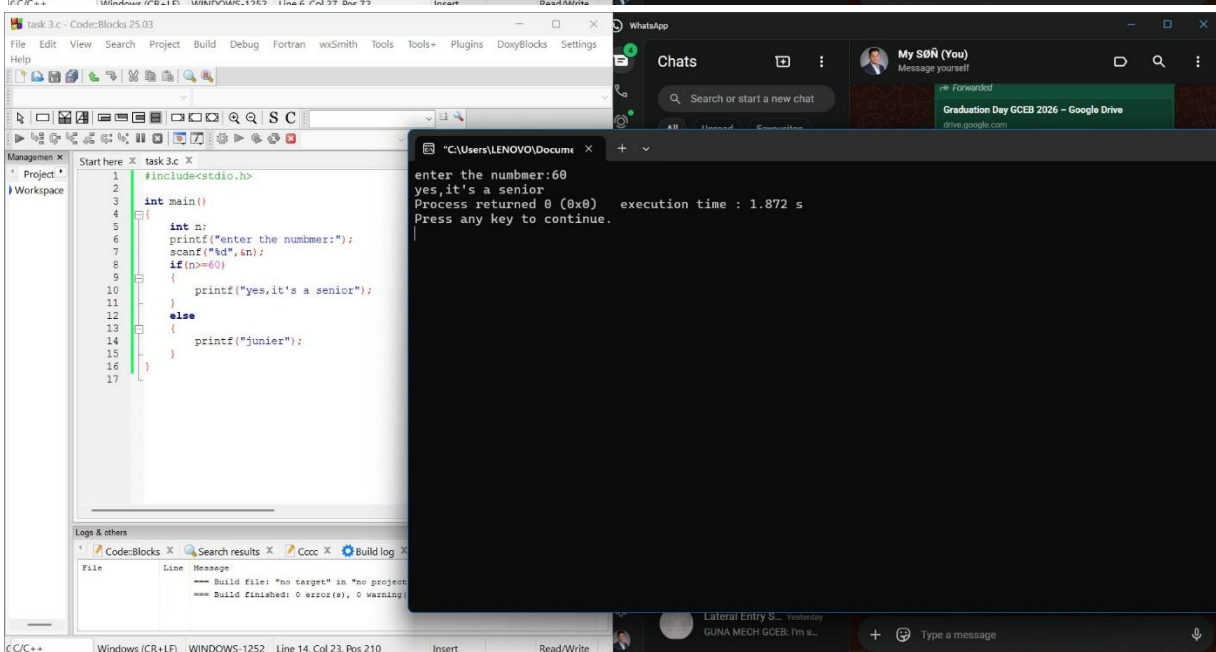
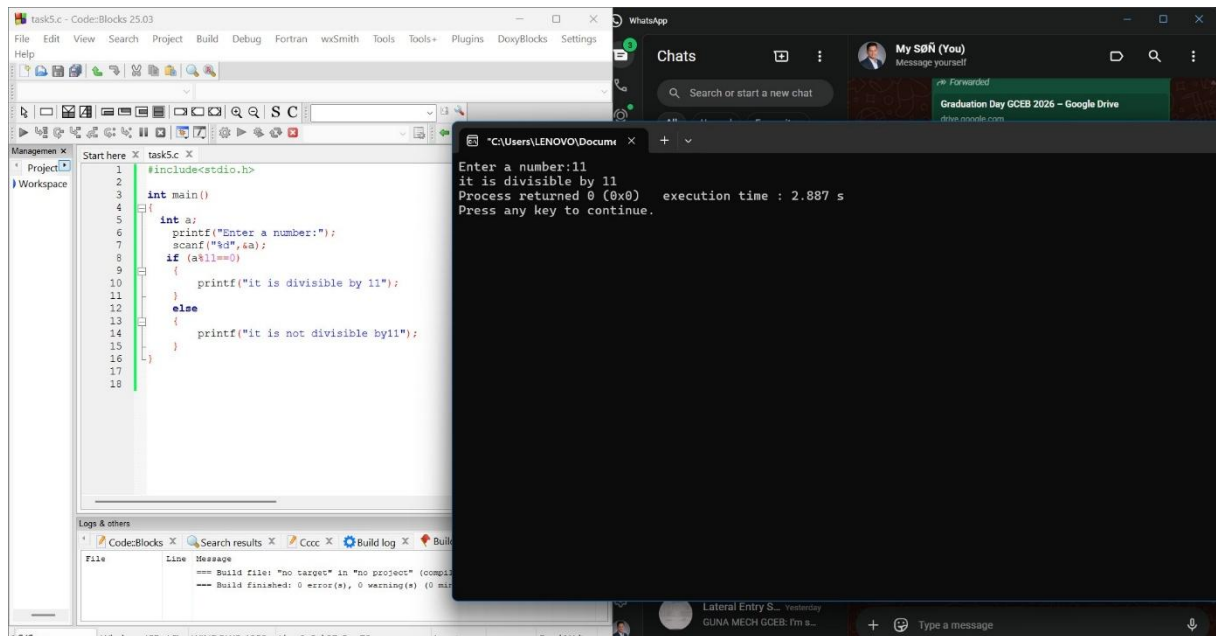
enter number:85
yes, it's two digit
Process returned 0 (0x0) execution time : 4.221 s
Press any key to continue.

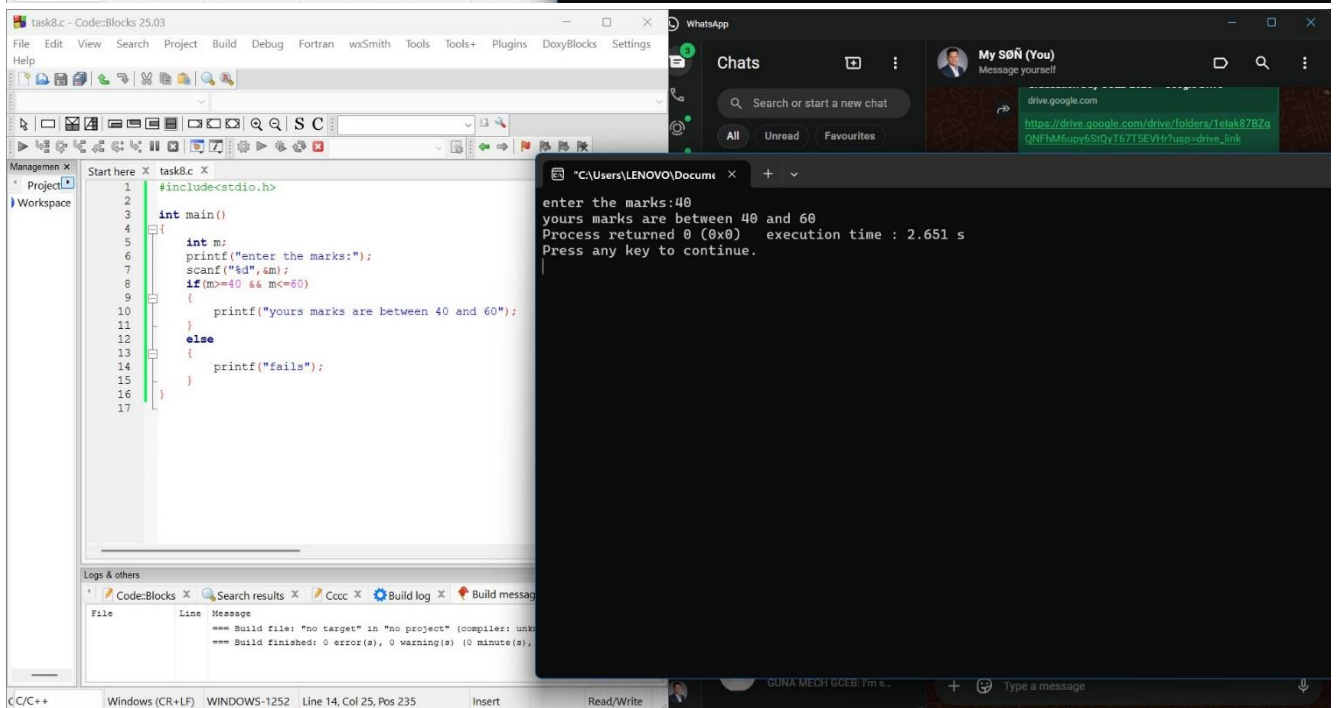
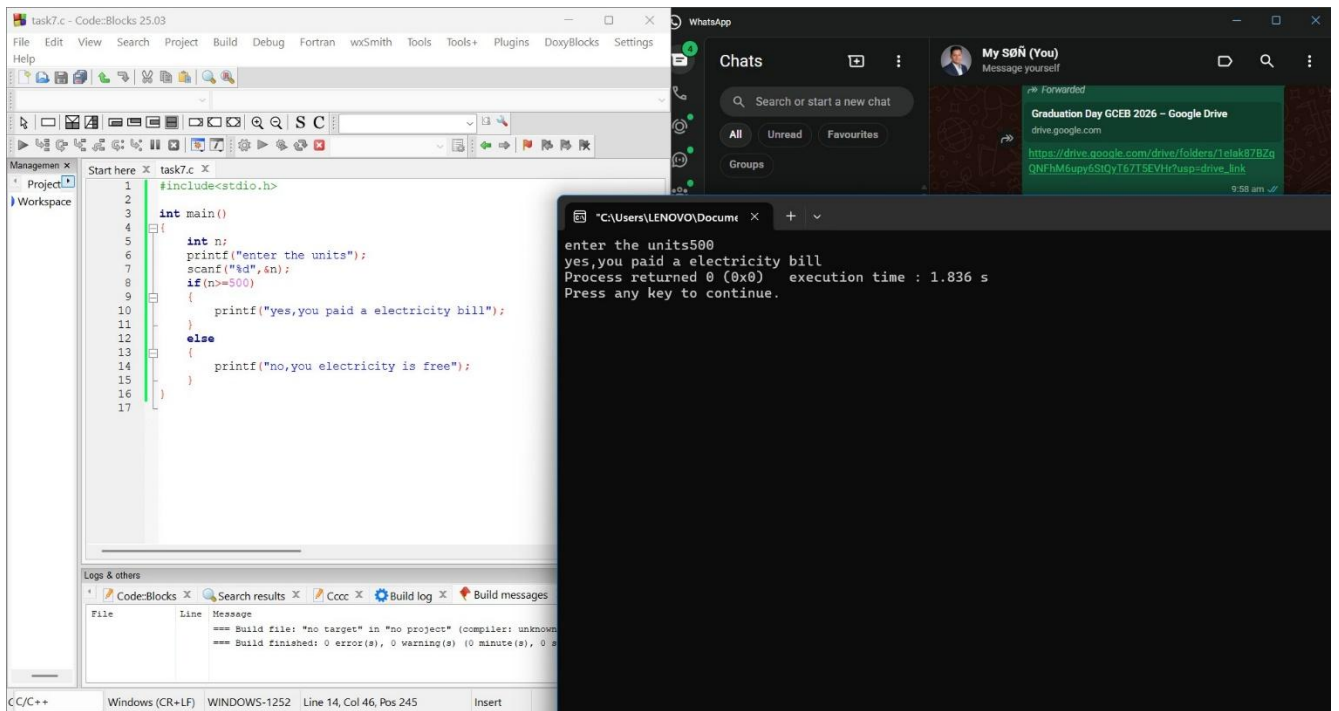


```
#include<stdio.h>

int main()
{
    int a;
    printf("enter number:");
    scanf("%d",&a);
    if(a%9==0)
    {
        printf("yes, divisible by 9");
    }
    else
    {
        printf("not");
    }
}
```

enter number:81
yes, divisible by 9
Process returned 0 (0x0) execution time : 3.773 s
Press any key to continue.





IF ELSE_IF ELSE:

The image displays three screenshots of a C++ IDE (Code::Blocks 25.03) showing different programs using conditional statements.

Top Screenshot: task1.c

```
1 #include <stdio.h>
2
3 int main() {
4     int month;
5
6     printf("Enter month number (1-12): ");
7     scanf("%d", &month);
8
9     if (month == 12 || month == 1 || month == 2) {
10        printf("Season: Winter\n");
11    }
12    else if (month == 3 || month == 4 || month == 5) {
13        printf("Season: Spring\n");
14    }
15    else if (month == 6 || month == 7 || month == 8) {
16        printf("Season: Summer\n");
17    }
18    else if (month == 9 || month == 10 || month == 11) {
19        printf("Season: Autumn\n");
20    }
21    else {
22        printf("Invalid month number! Please enter between 1 and 12.\n");
23    }
24 }
```

Output:

```
Enter month number (1-12): 6
Season: Summer
Process returned 0 (0x0)   execution time : 2.716 s
Press any key to continue.
```

Middle Screenshot: task3.c

```
1 #include <stdio.h>
2
3 int main() {
4     int balance = 0; // initial balance
5     int choice, amount;
6
7     while (1) {
8         printf("\n--- ATM Menu ---\n");
9         printf("1. Check Balance\n");
10        printf("2. Deposit\n");
11        printf("3. Withdraw\n");
12        printf("4. Exit\n");
13        printf("Enter your choice: ");
14        scanf("%d", &choice);
15
16        if (choice == 1) {
17            printf("Current Balance: %d\n", balance);
18        }
19        else if (choice == 2) {
20            printf("Enter deposit amount: ");
21            scanf("%d", &amount);
22            if (amount > 0) {
23                balance += amount;
24                printf("Deposited: %d, New Balance: %d\n", amount, balance);
25            }
26            else {
27                printf("Deposit amount must be positive!\n");
28            }
29        }
30        else if (choice == 3) {
31            printf("Enter withdrawal amount: ");
32        }
33    }
34 }
```

Output:

```
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 2
Enter deposit amount: 15244
Deposited: 15244, New Balance: 15244

--- ATM Menu ---
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 3500
Invalid choice! Try again.

--- ATM Menu ---
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 1
Current Balance: 15244

--- ATM Menu ---
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice:
```

Bottom Screenshot: task2.c

```
1 #include <stdio.h>
2
3 int main() {
4     int temp;
5
6     printf("Enter temperature in Celsius: ");
7     scanf("%d", &temp);
8
9     if (temp <= 0) {
10        printf("Temperature Category: Freezing\n");
11    }
12    else if (temp > 0 && temp <= 15) {
13        printf("Temperature Category: Cold\n");
14    }
15    else if (temp > 15 && temp <= 30) {
16        printf("Temperature Category: Warm\n");
17    }
18    else {
19        printf("Temperature Category: Hot\n");
20    }
21
22    return 0;
23 }
```

Output:

```
Enter temperature in Celsius: 25
Temperature Category: Warm
Process returned 0 (0x0)   execution time : 2.654 s
Press any key to continue.
```

The screenshot shows the Code::Blocks IDE with a C++ project named 'task4.c'. The code defines a function `main()` that prompts the user to enter a number and then checks its range using a series of `if-else` statements. The execution window shows the program running with the input '100', resulting in the output: 'the given number range is (51-100)'. The process returned 0 (0x0) and the execution time was 1.493 s.

```
#include<stdio.h>

int main()
{
    int num;
    printf("enter the number:");
    scanf("%d",&num);
    if(num>=0 && num<=10)
    {
        printf("the given number range is (0-10)");
    }
    else if(num>=11 && num<=50)
    {
        printf("the given number range is (11-50)");
    }
    else if(num>=51 && num<=100)
    {
        printf("the given number range is (51-100)");
    }
    else
    {
        printf("the given number is greater then 100");
    }
}
```

enter the number:100
the given number range is(51-100)
Process returned 0 (0x0) execution time : 1.493 s
Press any key to continue.

The screenshot shows the Code::Blocks IDE with a C++ project named 'task5.c'. The code defines a function `main()` that prompts the user to enter a number and then checks if it is positive, negative, or zero using `if-else` statements. The execution window shows the program running with the input '-1', resulting in the output: 'number is negative'. The process returned 0 (0x0) and the execution time was 4.092 s.

```
int main()
{
    int num;
    printf("enter the number:");
    scanf("%d",&num);
    if(num>0)
    {
        printf("number is positive");
    }
    else if(num<0)
    {
        printf("number is nagtive");
    }
    else
    {
        printf("number is zero");
    }
}
```

enter the number:-1
number is negative
Process returned 0 (0x0) execution time : 4.092 s
Press any key to continue.

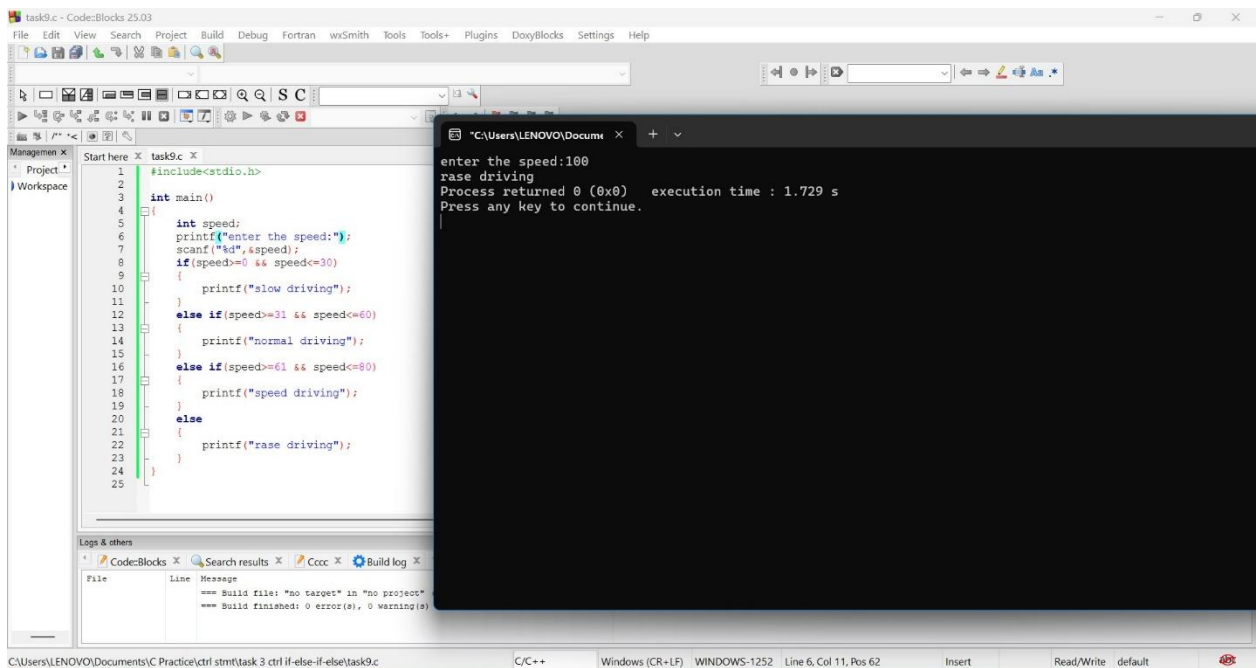
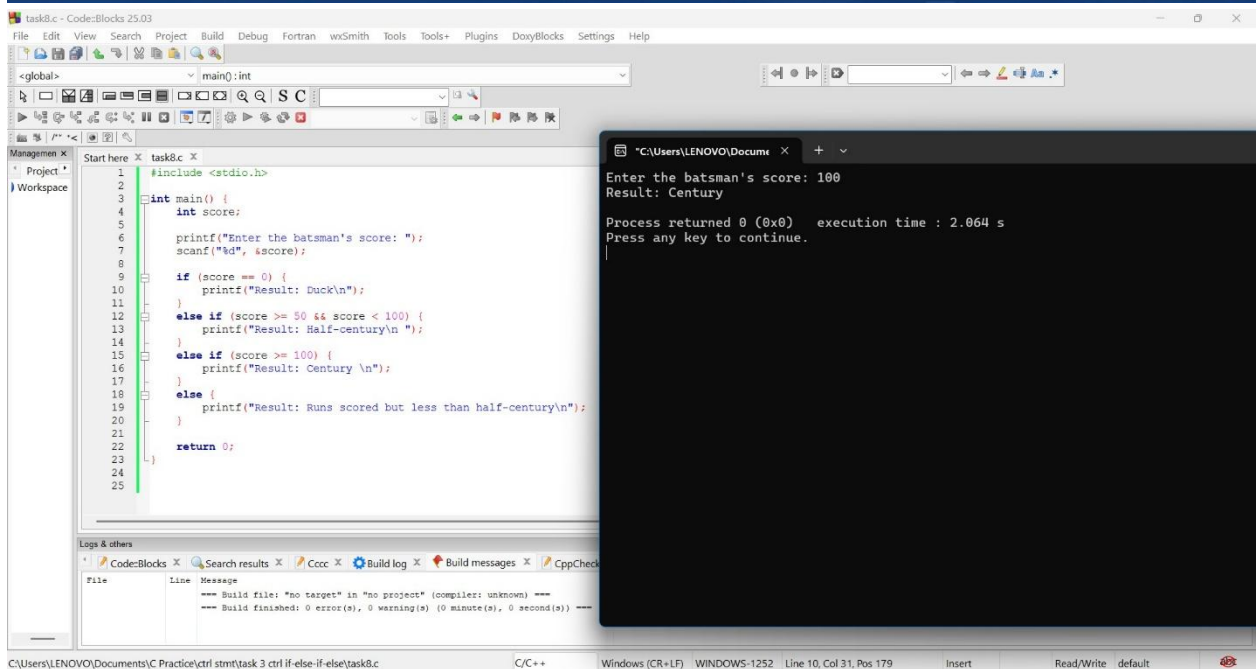
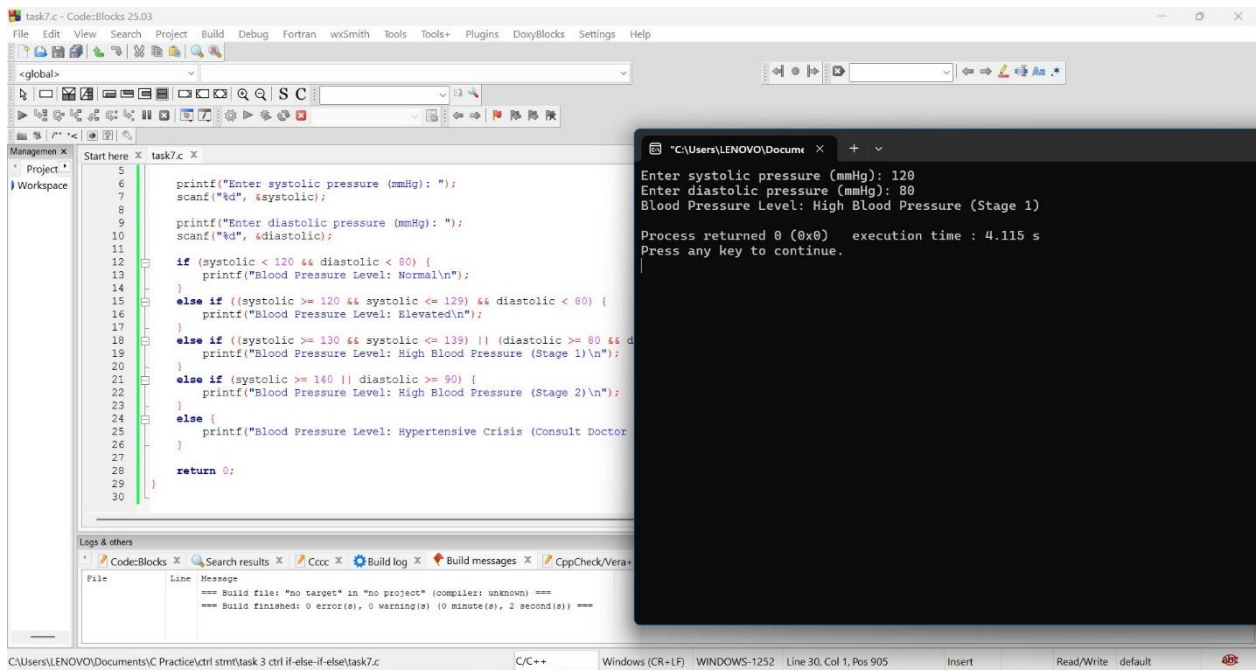
The screenshot shows the Code::Blocks IDE with a C++ project named 'task6.c'. The code defines a function `main()` that prompts the user to enter student marks and annual family income, then checks eligibility for a scholarship based on these values using `if-else` statements. The execution window shows the program running with the input '50' for marks and '10000' for income, resulting in the output: 'Eligible for MERIT Scholarship'. The process returned 0 (0x0) and the execution time was 11.528 s.

```
int main() {
    int marks;
    float income;
    printf("Enter student marks (out of 100): ");
    scanf("%d", &marks);
    printf("Enter annual family income (in Rs): ");
    scanf("%f", &income);

    if (marks >= 75 && income <= 200000) {
        printf("Eligible for FULL Scholarship ");
    }
    else if (marks >= 60 && income <= 500000) {
        printf("Eligible for PARTIAL Scholarship ");
    }
    else if (marks >= 50 && income <= 800000) {
        printf("Eligible for MERIT Scholarship");
    }
    else {
        printf("Not Eligible for Scholarship ");
    }

    return 0;
}
```

Enter student marks (out of 100): 50
Enter annual family income (in Rs): 10000
Eligible for MERIT Scholarship
Process returned 0 (0x0) execution time : 11.528 s
Press any key to continue.



task10.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

<global>

Start here X task10.c X

```
1 int main() {
2     int wheels;
3
4     printf("Enter number of wheels: ");
5     scanf("%d", &wheels);
6
7     if (wheels == 2) {
8         printf("Transport Type: Bike\n");
9     }
10    else if (wheels == 4) {
11        printf("Transport Type: Car\n ");
12    }
13    else if (wheels > 4 && wheels <= 6) {
14        printf("Transport Type: Bus \n");
15    }
16    else if (wheels > 6) {
17        printf("Transport type: Truck \n");
18    }
19    else {
20        printf("Invalid input! Please enter a valid wheel c
21    }
22    return 0;
23 }
24
25
26
27
```

Logs & others

Code::Blocks X Search results X Cccc X Build log X Build messages X

File Line Message

--- Build file: "no target" in "no project" (compiler: unknown)

--- Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 sec

Enter number of wheels: 6
Transport Type: Bus

Process returned 0 (0x0) execution time : 1.316 s
Press any key to continue.

WhatsApp

Chrono Copy | Mohamed

This Tommy Hilfiger is a bold chronograph timepiece that perfectly blends sporty aesthetics with refined elegance, making it ideal for both eve

556 unread messages

C:\Users\LENOVO\Documents\C Practice\ctrl stmt\task 3 ctrl if-else-if-else\task10.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 19, Col 41, Pos 439 Insert Ready Write default

task11.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

<global>

Start here X task11.c X

```
1 int main() {
2     int years;
3
4     printf("Enter years of experience: ");
5     scanf("%d", &years);
6
7     if (years < 1) {
8         printf("Bonus: 0% (No bonus)\n");
9     }
10    else if (years >= 1 && years < 3) {
11        printf("Bonus: 5%\n");
12    }
13    else if (years >= 3 && years < 5) {
14        printf("Bonus: 10%\n");
15    }
16    else if (years >= 5 && years < 10) {
17        printf("Bonus: 15%\n");
18    }
19    else {
20        printf("Bonus: 20%\n");
21    }
22    return 0;
23 }
24
25
26
27
28
```

Logs & others

Code::Blocks X Search results X Cccc X Build log X Build me

File Line Message

--- Build file: "no target" in "no project" (compiler:)

--- Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 sec

Enter years of experience: 10
Bonus: 20%

Process returned 0 (0x0) execution time : 5.859 s
Press any key to continue.

C:\Users\LENOVO\Documents\C Practice\ctrl stmt\task 3 ctrl if-else-if-else\task11.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 27, Col 1, Pos 518 Insert Read/Write default

task12.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

<global>

Start here X task12.c X

```
1 int main() {
2     int marks;
3
4     printf("Enter total marks (out of 100): ");
5     scanf("%d", &marks);
6
7     if (marks < 40) {
8         printf("Result: Fail \n");
9     }
10    else if (marks >= 40 && marks < 50) {
11        printf("Result: Pass \n");
12    }
13    else if (marks >= 50 && marks < 60) {
14        printf("Result: Merit \n");
15    }
16    else if (marks >= 60 && marks < 75) {
17        printf("Result: Distinction \n");
18    }
19    else {
20        printf("Result: Topper \n");
21    }
22    return 0;
23 }
24
25
26
27
28
```

Logs & others

Code::Blocks X Search results X Cccc X Build log X Build messages

File Line Message

--- Build file: "no target" in "no project" (compiler: unknown)

--- Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 s

Enter total marks (out of 100): 95
Result: Topper

Process returned 0 (0x0) execution time : 87.828 s
Press any key to continue.

C:\Users\LENOVO\Documents\C Practice\ctrl stmt\task 3 ctrl if-else-if-else\task12.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 22, Col 32, Pos 506 Insert Read/Write default

NESTED_IF:

The screenshot shows the Code::Blocks IDE with a project named 'task1.c'. The source code is as follows:

```
1 #include <stdio.h>
2
3 int main() {
4     int num;
5     printf("Enter a number: ");
6     scanf("%d", &num);
7
8     if (num > 0) { // First check: positive
9         printf("The number is positive.\n");
10
11         if (num % 2 == 0) { // Nested check: even or odd
12             printf("It is even.\n");
13         } else {
14             printf("It is odd.\n");
15         }
16     } else {
17         printf("The number is not positive.\n");
18     }
19
20     return 0;
21 }
```

The execution output in the terminal window is:

```
Enter a number: 254
The number is positive.
It is even.

Process returned 0 (0x0)   execution time : 17.856 s
Press any key to continue.
```

The screenshot shows the Code::Blocks IDE with a project named 'task2.c'. The source code is as follows:

```
16 printf("Enter mark for Subject 4: ");
17 scanf("%d", &sub4);
18
19 printf("Enter mark for Subject 5: ");
20 scanf("%d", &sub5);
21
22 avg = (sub1 + sub2 + sub3 + sub4 + sub5) / 5.0;
23
24 // First check: passed all subjects (pass mark = 40)
25 if (sub1 >= 40 && sub2 >= 40 && sub3 >= 40 && sub4 >= 40 && sub5 >= 40) {
26     printf("Student has passed all subjects.\n");
27
28     // Nested check: distinction eligibility
29     if (avg >= 75) {
30         printf("Eligible for distinction.\n");
31     } else {
32         printf("Not eligible for distinction.\n");
33     }
34 } else {
35     printf("Student has failed in one or more subjects.\n");
36 }
37
38 return 0;
39
40
41 }
```

The execution output in the terminal window is:

```
Enter mark for Subject 1: 95
Enter mark for Subject 2: 90
Enter mark for Subject 3: 89
Enter mark for Subject 4: 99
Enter mark for Subject 5: 94
Student has passed all subjects.
Eligible for distinction.

Process returned 0 (0x0)   execution time : 10.171 s
Press any key to continue.
```

The screenshot shows the Code::Blocks IDE with a project named 'task3.c'. The source code is as follows:

```
3 int main() {
4     int age;
5     char license;
6
7     printf("Enter your age: ");
8     scanf("%d", &age);
9
10    if (age >= 18) { // First check: age eligibility
11        printf("You are eligible by age.\n");
12
13        printf("Do you have a driving license? (y/n): ");
14        scanf(" %c", &license); // space before %c to handle newline
15
16        if (license == 'y' || license == 'Y') { // Nested check
17            printf("You are allowed to drive.\n");
18        } else {
19            printf("You must obtain a driving license first.\n");
20        }
21    } else {
22        printf("You are not eligible by age.\n");
23    }
24
25    return 0;
26
27
28 }
```

The execution output in the terminal window is:

```
Enter your age: 18
You are eligible by age.
Do you have a driving license? (y/n): y
You are allowed to drive.

Process returned 0 (0x0)   execution time : 6.375 s
Press any key to continue.
```

task4.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

Start here X task4.c X

```
6 // Predefined correct credentials
7 char correct_username[] = "admin";
8 char correct_password[] = "12345";
9
10 printf("Enter username: ");
11 scanf("%s", username);
12
13 if (strcmp(username, correct_username) == 0) { // First check
14     printf("Username is correct.\n");
15
16     printf("Enter password: ");
17     scanf("%s", password);
18
19     if (strcmp(password, correct_password) == 0) { // Nested
20         printf("Login successful!\n");
21     } else {
22         printf("Incorrect password.\n");
23     }
24 } else {
25     printf("Incorrect username.\n");
26 }
27
28 return 0;
```

Logs & others

Code::Blocks X Search results X Ccc X Build log X Build messages X CppC

File Line Message

--- Build file: "no target" in "no project" (compiler: unknown) ---
--- Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ---

C:\Users\LENOVO\Documents\C Practice\ctrl stmt\task 4 ctrl stmt nested if\task4.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 31, Col 1, Pos 762 Insert Read/Write default

Enter username: admin
Username is correct.
Enter password: 12345
Login successful!

Process returned 0 (0x0) execution time : 7.347 s
Press any key to continue.

task5.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

<global>

Start here X task5.c X

```
1 #include <stdio.h>
2
3 int main() {
4     int num;
5
6     printf("Enter a number: ");
7     scanf("%d", &num);
8
9     // First check: divisible by 5
10    if (num % 5 == 0) {
11        printf("The number is divisible by 5.\n");
12
13        // Nested check: divisible by 10
14        if (num % 10 == 0) {
15            printf("It is also divisible by 10.\n");
16        } else {
17            printf("It is not divisible by 10.\n");
18        }
19    } else {
20        printf("The number is not divisible by 5.\n");
21    }
22
23    return 0;
```

Logs & others

Code::Blocks X Search results X Ccc X Build log X Build messages X CppC

File Line Message

--- Build file: "no target" in "no project" (compiler: unknown) ---
--- Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ---

C:\Users\LENOVO\Documents\C Practice\ctrl stmt\task 4 ctrl stmt nested if\task5.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 25, Col 1, Pos 531 Insert Read/Write default

Enter a number: 40
The number is divisible by 5.
It is also divisible by 10.

Process returned 0 (0x0) execution time : 1.556 s
Press any key to continue.

task6.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

Start here X task6.c X

```
4 int year;
5
6 printf("Enter a year: ");
7 scanf("%d", &year);
8
9 // First check: divisible by 4
10 if (year % 4 == 0) {
11     // Nested check: divisible by 100
12     if (year % 100 == 0) {
13         // Further nested check: divisible by 400
14         if (year % 400 == 0) {
15             printf("%d is a leap year.\n", year);
16         } else {
17             printf("%d is not a leap year.\n", year);
18         }
19     } else {
20         printf("%d is a leap year.\n", year);
21     }
22 } else {
23     printf("%d is not a leap year.\n", year);
24 }
25
26 return 0;
```

Logs & others

Code::Blocks X Search results X Ccc X Build log X Build messages X CppC

File Line Message

--- Build file: "no target" in "no project" (compiler: unknown) ---
--- Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ---

C:\Users\LENOVO\Documents\C Practice\ctrl stmt\task 4 ctrl stmt nested if\task6.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 28, Col 1, Pos 662 Insert Read/Write default

Enter a year: 2026
2026 is not a leap year.

Process returned 0 (0x0) execution time : 2.732 s
Press any key to continue.

The screenshot shows the Code::Blocks IDE with a project named 'task7.c'. The source code is as follows:

```
1 // task7.c
2
3 #include <stdio.h>
4
5 int main() {
6     float salary;
7     int experience;
8
9     printf("Enter your salary: ");
10    scanf("%f", &salary);
11
12    if (salary > 30000) { // First check: salary
13        printf("Salary is above 30,000.\n");
14
15        printf("Enter your years of experience: ");
16        scanf("%d", &experience);
17
18        if (experience >= 5) { // Nested check: experience
19            printf("Eligible for bonus.\n");
20        } else {
21            printf("Not eligible for bonus (experience <= 5 years).\n");
22        }
23    } else {
24        printf("Salary is not above 30,000.\n");
25    }
26
27    return 0;
28 }
```

The execution output is shown in a terminal window:

```
Enter your salary: 50000
Salary is above 30,000.
Enter your years of experience: 5
Eligible for bonus.

Process returned 0 (0x0)   execution time : 7.536 s
Press any key to continue.
```

The screenshot shows the Code::Blocks IDE with a project named 'task8.c'. The source code is as follows:

```
1 // task8.c
2
3 #include <stdio.h>
4
5 int main() {
6     printf("Enter a number: ");
7     scanf("%d", &num);
8
9     // First check: three-digit number
10    if (num >= 100 && num <= 999) {
11        printf("The number is three-digit.\n");
12
13        // Extract digits
14        d1 = num / 100; // first digit
15        d2 = (num / 10) % 10; // middle digit
16        d3 = num % 10; // last digit
17
18        // Nested check: palindrome
19        if (d1 == d3) {
20            printf("It is a palindrome.\n");
21        } else {
22            printf("It is not a palindrome.\n");
23        }
24    } else {
25        printf("The number is not three-digit.\n");
26    }
27
28    return 0;
29 }
```

The execution output is shown in a terminal window:

```
Enter a number: 444
The number is three-digit.
It is a palindrome.

Process returned 0 (0x0)   execution time : 2.140 s
Press any key to continue.
```

The screenshot shows the Code::Blocks IDE with a project named 'task9.c'. The source code is as follows:

```
1 // task9.c
2
3 #include <stdio.h>
4
5 int main() {
6     char ch;
7
8     printf("Enter a character: ");
9     scanf("%c", &ch);
10
11    // First check: is it an alphabet?
12    if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
13        printf("It is an alphabet.\n");
14
15        // Nested check: uppercase or lowercase
16        if (ch >= 'A' && ch <= 'Z') {
17            printf("It is uppercase.\n");
18        } else {
19            printf("It is lowercase.\n");
20        }
21    } else {
22        printf("It is not an alphabet.\n");
23    }
24
25    return 0;
26 }
```

The execution output is shown in a terminal window:

```
Enter a character: a
It is an alphabet.
It is lowercase.

Process returned 0 (0x0)   execution time : 3.423 s
Press any key to continue.
```